

Armouring Up

The centre is looking to put the deployment of the Kavach system on a fast track to strengthen the security of the Railways—the lifeline of the nation

BY RICHA SHARMA

22 | ► ON JUNE 17, a goods train rammed into the Kanchanjunga Express in West Bengal, killing 10 and injuring many. The accident has renewed calls for expediting the deployment of Kavach, an automatic train protection (ATP) system which is so far operational in just 2% of the 69,000-km long railway network.

The system currently covers 1,465 km and 139 locomotives (including Electric Multiple Unit rakes) on the South Central Railway network. A further 6,000 km has been tendered and the tendering process for another 10,000 km is underway.

Kavach aids the loco pilot in trains running within specified speed limits by automatically applying brakes in case the loco pilot fails and also helps the train run safely during harsh weather.

Other benefits include controlling speed of trains by automatic application of brakes on the approach of turnouts, repeating of signal aspects in the cab, useful for higher speeds and foggy weather, and auto whistles at level crossing gates.

The Union government earmarked ₹710 crore for the system in the FY24 Budget and another ₹560 crore was allocated in the FY25 Interim Budget. Fitting a train with Kavach technology costs around ₹70 lakh and the trackside and station equipment cost around ₹50 lakh, per kilometer.

Currently, Kavach contracts have been awarded for Delhi-Mumbai (including Ahmedabad-Vadodara section) & Delhi-Howrah (including Lucknow-Kanpur section) corridors approximately covering 3,000 km under Eastern Railway, East Central Railway, North Central Railway, Northern Railway, West Central Railway, and Western Railway.

So, why is the rollout taking time?

Railway Minister Ashwini Vaishnaw said Kavach has five components, which need to be customised for different trains.

“For example, a 5,000-tonne goods train would need a 1-1.5 km distance to come to a halt while for mainline electric multiple unit (MEMU), it will be 300-500 metres. So, the system has to be customised for different train categories,” he said at an informal interaction after taking charge as Minister of Railways.

The ministry is looking to accelerate the implementation of the system to cover 44,000 km in the next five years. Vaishnaw chaired a meeting on Kavach 4.0, asking officials to speed up the process.

Sharing the progress of implementation, Vaishnaw said optical fibre cables had been laid on 3,040 km by 2023-end and trackside equipment has been installed on 827 km.

Experts also point at organisational weaknesses at the ground level that need to be fixed.

SAFETY MATH (2014-2024)

₹1.78

lakh crore

SPENT ON SAFETY-RELATED WORKS

₹1.09

lakh crore

SPENT ON TRACK RENEWAL

₹36,699

crore

SPENT ON LEVEL-CROSSING ELIMINATION

₹8,000

crore

SPENT ON RAILWAY BRIDGE REHABILITATION

₹45,000

crore

SPENT ON RASHTRIYA RAIL SANRAKSHA KOSH

SOURCE MINISTRY OF RAILWAYS